

# WASP-43b

R-1963

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-43\_210108 · created 2026-08-06T23:09:50+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2459222.8518 +/- 0.0019 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.1 +/- 0.049                   |
| Transit depth (Rp/Rs)^2                           | 0.99 +/- 0.97 [%]               |
| Orbital Inclination (inc)                         | 82.24 +/- 2.82                  |
| Airmass coefficient 1 (a1)                        | 1.013 +/- 0.025                 |
| Airmass coefficient 2 (a2)                        | -0.009 +/- 0.015                |
| Scatter in the residuals of the lightcurve fit is | 2.45 %                          |
| Best Comparison Star                              | #1 - [288, 36]                  |
| Optimal Method                                    | PSF photometry                  |
| Transit Duration (day)                            | 0.045 +/- 0.012                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                         |
|---------------------------|-----------------------------------------|
| Rp/R* fit vs published    | 0.1 ± 0.049 vs 0.1594 (deviation 1.21σ) |
| Duration fit vs published | 0.045 d vs 0.0512 d (deviation 0.52σ)   |
| Mid-time O-C              | -3.8 min                                |
| Depth SNR                 | 2.0                                     |
| Residual scatter          | 2.45 %                                  |

### Light curve

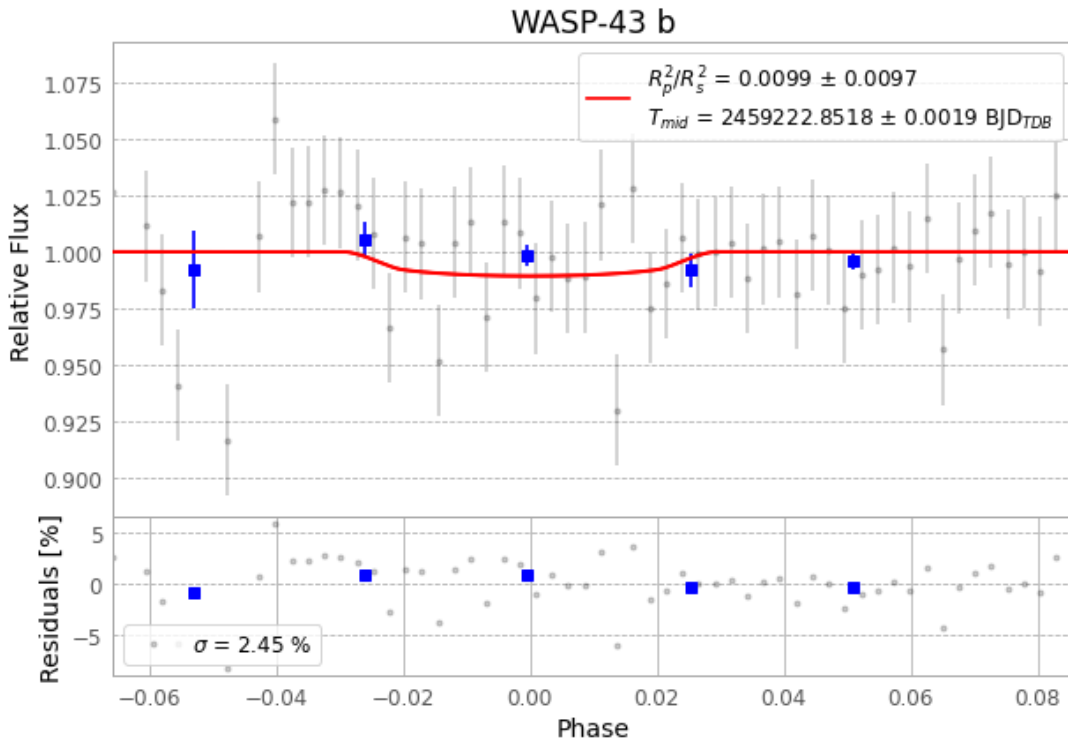


Figure 1 — FinalLightCurve\_WASP-43 b\_2021-01-08.png (SHA-256 17ec07736152cff2...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [363, 231], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 e7b1396a212e65d2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit fd74a91

## Data provenance

59 FITS frames (39 MB), WASP-43210108070314.FITS → WASP-43210108100012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-210108.log (e2952b8cbc02...), FinalLightCurve\_WASP-43 b\_2021-01-08.png (17ec07736152...), FinalLightCurve\_WASP-43 b\_2021-01-08.csv (8ebe6e5e3382...)

## Compute resources

Wall time 361.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 23:09Z.